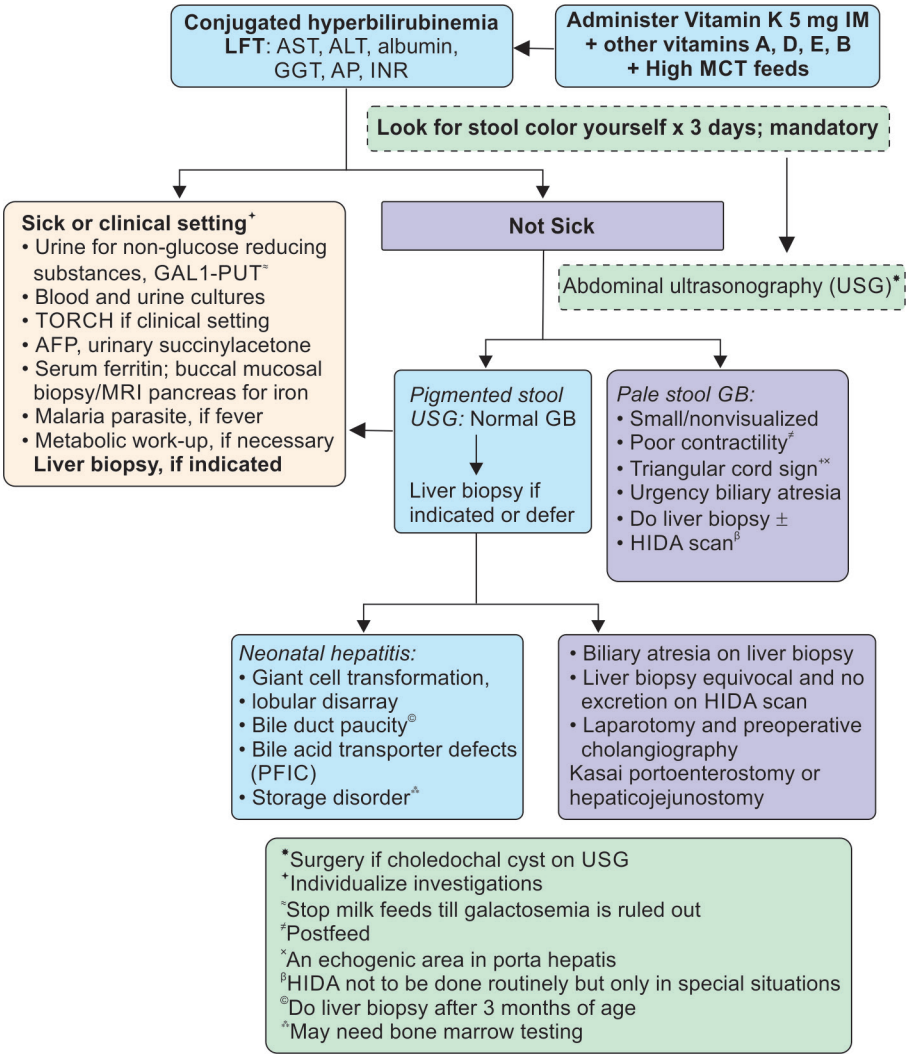
Approach to a Case of Neonatal Cholestasis



(AFP: alpha-fetoprotein; ALT: alanine transaminase; AP: alkaline phosphatase; AST: aspartate transaminase; GAL1-PUT: galactose-1-phosphate uridyltransferase; GB: gallbladder; GGT: gamma-glutamyl transferase; HIDA: hepatobiliary iminodiacetic acid; INR: international normalized ratio; LFT: liver function test; MCT: medium chain triglycerides; PFIC: progressive familial intrahepatic cholestasis)

Disease entity	Specific investigations		Specific treatment
	Screening	Confirmatory	
Biliary atresia (BA)	Pale-colored stool, high GGT, USG: GB rudimentary/poor contractility, triangular cord sign+; liver biopsy features of BA	Peroperative cholangiogram	☑ *Surgical:* Portoenterostomy in the large majority of cases Or ☑ Hepaticojejunostomy in a few with patent confluence at porta
Galactosemia	Urine for nonglucose reducing substances +	Galactose-1-phosphate uridyltransferase (GAL1-PUT)	Lactose-free diet
Tyrosinemia	High AFP	Urinary succinylacetone present (being highly volatile substance sample should be quickly analysis)	Drug treatment with NTBC [2-(2-nitro-4-trifluoromethylbenzoyl)-1, 3-cyclohexanedione] and restricted tyrosine and phenylalanine diet Alternatively liver transplantation
Gestational alloimmune liver disease (earlier named as neonatal hemochromatosis)	High serum ferritin median 2,448 (415–100,000) µg/L	*Lip biopsy (submucosal gland):* Iron deposition on staining Or MRI pancreas for iron deposition	High dose of intravenous immunoglobulin ± exchange blood transfusions. If no response, then liver transplantation
Bile acid transporter defects (PFIC types 1-VI; types I-III being common)	☑ *GGT:* Normal/low, high ☑ Liver biopsy features	Genetic evaluation (clinical exome sequencing)	Medical treatment for pruritus. May need surgical biliary diversion to control pruritus. Liver transplantation, if indicated
Alagille syndrome	High GGT; high triglycerides; eye, cardiac, and skeletal specified defects; paucity of bile ducts on liver biopsy	Genetic evaluation JAG 1 or NOTCH 2 mutation	Same as that of bile acid transporter defects with some exceptions in managing disease complications

Drug/supplementation	*Dose*	*Side effects/comment*
Vitamin K	2.5 mg twice/week to 5 mg/day oral Or IV/IM: 2–5 mg/month	None *(To be given if coagulopathy is present or till conjugated jaundice is existing)*
Vitamin D (cholecalciferol)	Oral 2,500–4,000 IU/day Or One sachet (60,000 IU) per month	Hypercalcemia, nephrocalcinosis *(To be given if coagulopathy is present or till conjugated jaundice is existing). May monitor vitamin D levels every 3 monthly and treat accordingly).*
Vitamin A	Oral 5,000–25,000 IU/day	Hepatotoxicity, hypercalcemia Pseudotumor cerebri if higher dose is used. Avoid hypervitaminosis as it can enhance liver fibrosis
Vitamin E	Oral 50–400 IU/day	Potentiation of vitamin K deficiency coagulopathy, diarrhea
Water-soluble vitamins	Twice the recommended daily allowances	None
Calcium supplementation	Oral 25–100 mg/kg/day	
Medium chain triglycerides (MCT)	15–20% of calories	Oral supplementation in the form of oil/powder/formula feeds containing desired MCT content
Pruritus (oral drugs)¤		
Ursodeoxycholic acid	10–20 mg/kg/day	Diarrhea, hepatotoxicity
Rifampicin	10 mg/kg/day	Hepatotoxicity, drug interactions
Phenobarbitone	3–10 mg/kg/day	Sedative effect, behavioral changes
Cholestyramine	0.25–0.5 g/kg/day	Constipation, steatorrhea, hyperchloremic metabolic acidosis

Note: Vitamins to be administered in all cases of NC irrespective of etiology. Fat- and water-soluble vitamins to be given till conjugated jaundice is present otherwise when indicated depending upon the disease status.

¤ Recently newer drugs have been tried as ileal (intestinal) bile acid transport inhibitors for treating cholestatic liver disease

Further Reading

- ☑ Bhatia V, Bavdekar A, Matthai J, Waikar Y, Sibal A. Management of Neonatal Cholestasis: Consensus Statement of the Pediatric Gastroenterology Chapter of Indian Academy of Pediatrics. Indian Pediatr. 2014;51:203-10.
- ☑ Feldman AG, Sokol RJ. Recent developments in diagnostics and treatment of neonatal cholestasis. Semin Pediatr Surg. 2020;29:150945.
- ☑ Indian Academy of Pediatrics. Consensus Report on Neonatal Cholestasis Syndrome. Pediatric Gastroenterology Subspecialty Chapter of Indian Academy of Pediatrics Document. Indian Pediatr. 2000;37:845-51.

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Neonatal Resuscitation Program

Lead Author

Lalan K Bharti

Co-Authors

A Prakash, Manish Jain

Under the Auspices of the IAP Action Plan 2022

Remesh Kumar R
IAP President 2022

Upendra Kinjawadekar
IAP President-Elect 2022

Piyush Gupta
IAP President 2021

Vineet Saxena
IAP HSG 2022–2023

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IAP Standard Treatment Guidelines Committee

Chairperson

Remesh Kumar R

IAP Coordinator

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